Fiscal Year:	FY 2020	**Task Last Updated:**	FY 02/24/2020
PI Name:	Law, Chung K. Ph.D.		
Project Title:	Structure and Response of Spherical Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Chung Law's email is cklaw@princeton.edu		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	OTHER
Start Date:	11/25/2014	**End Date:**	08/26/2021
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 8/26/2021 due to CoInvestigator Stephen Tse's accompanying grant that ends 8/26/2021 (Ed., 4/20/2020) NOTE: Extended to 11/24/2019 per NSSC info (Ed., 11/25/16)		
Key Personnel Changes/Previous PI:	NOTE April 2020: Stephen Tse, CoInvestigator, continues work under this grant with his grant 80NSSC18M0134, which continues through 8/26/2021; he will continue reporting.		
COI Name (Institution):	Tse, Stephen Ph.D. (Rutgers University grant 80NSSC18M0134)		
Grant/Contract No.:	NNX15AC79A ; 80NSSC18M0134 (CoI Tse)		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of Structure and Response of Spherical Diffusion Flames--NNX09AR03A with the same Principal Investigator, Chung K. Law. NOTE April 2020: Stephen Tse, CoInvestigator, continues work under this grant with his grant 80NSSC18M0134, which continues through 8/26/2021; CoI Tse will continue reporting. The effort was divided into experimental and theoretical elements. The experimental effort consisted of using a porous spherical burner to issue fuel mixtures into an oxidizing environment. Different mixtures of fuel species permitted investigation of the phenomena of interest. The experiment effort was conducted primarily in the drop towers at the Glenn Research Center (GRC) and the NASA reduced-gravity aircraft. Space flight experiment was fully defined and approved during the program period. The theoretical effort consisted of the development of computational codes simulating the combustion phenomena of interest, and interpreting the calculated results on the basis of fundamental flame physics. The purpose of this flight definition task is to gain fundamental understanding of the structure and dynamics of diffusion flames in a simple, well-defined flow field. Using a spherical geometry this work considers: 1) gas-phase transient processes, including fuel vapor accumulation inside the flame and far-field diffusion, and their influence on flame movement and the achievement of steady flame behavior; 2) extinction boundaries of steady or near-steady flames and the identification of the associated chemistry at the global, reduced, and detailed levels; 3) the characteristics of partially-premixed flames; 4) flamefront pulsation and cellular instability especially near the extinction limits; 5) the dynamics of rotating flames; and 6) the behavior and structure of quasi-spherical flames in partial gravity environments. These studies are intended to significantly enhance the understanding of diffusion flames, especially providing capability for predicting flame behavior under varied conditions and configurations. As such, this work should contribute to engineering approaches to practical combustion systems, achieving increased energy conversion efficiencies and reductions in unwanted combustion emissions in both terrestrial and extraterrestrial environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enhanced understanding of flame-flow vortical interaction could lead to improved burner design and operation in terms of energy efficiency and pollutant emissions.
	During the reporting period, results from the following investigations have been obtained. Initiation and Propagation of Laminar Premixed Cool Fames. Initiation and propagation of premixed cool flames for DME/O2/N2 mixtures are computationally studied in the 1-D planar configuration as well as the counterflow configuration. The results demonstrate that a residence time window exists for the occurrence of cool flames; external to this window a cool flame either cannot be initiated or will transition to a hot flame. Parametric study shows that the cool flame regime is sensitive to the amount of dilution in the flow, and that the cool flame becomes more intense at richer conditions. Global parameters such as the reference cool flame speed and cool flame temperature have been studied in a symmetric counterflow for various strain rates and equivalence ratios, with the controlling reactions identified. The results show that the premixed cool flame speeds are relatively low, around 6-7 cm/s, and are much less sensitive in terms of equivalence ratio variations. Experimental investigations have also been carried out in a counterflow, with a steady cool flame observed with a CCD camera with high sensitivity of the UV band. Such observations qualitatively demonstrate stronger cool flame chemiluminescence and insensitive change in the flame position with increasing equivalence ratios, which correspond well with the detailed numerical simulation. This work is reported in: “Initiation and propagation of laminar premixed cool flames,” Peng Zhao, Wenkai Liang, Sili Deng, and Chung K. Law, Fuel 166 (2016) 477-487. [Ed. note: see Cumulative Bibliography] Aspects of Colliding Spherical Flames: The morphology, corner dynamics, and flame-generated vorticity in colliding spherical flames are investigated. Schlieren images of the flame surface show that the morphology of the colliding process can be divided into three stages, which is further supported by simulations. Using experimental and simulation results, a comprehensive understanding of the collision dynamics is acquired, particularly the corner dynamics and the vortices generated near the corner. Regarding the former, we have identified strongly interacting factors, namely preferential diffusion and kine- matic restoration. Regarding the latter, we have demonstrated that the vorticity is generated by the baroclinic torque across the flame as a result of hydrodynamic pressure redistribution and density jump. The magnitude of the vorticity is estimated under quasi-steadiness assumption. By comparing the scales of the kinetic energy of the flame-generated vortices with those of the Kolmogorov eddies, it is suggested that vorticity at the flame corner could persist over dissipation and accumulate to a substantial extent to affect the turbulence spectrum as well as the local/global burning rate. This work is reported in: “On Colliding Spherical Flames: Morphology, Corner Dynamics, and Flame-generated Vorticity,” Sheng Yang, Delin Zhu, and Chung K. Law, Combustion and Flame 167 (2016) 444-451. [Ed. note: see Cumulative Bibliography] Extended Flammability Limits of n-Heptane/Air Mixtures with Cool Flames: A numerical and theoretical investigation has been conducted on the flammability limits of n-heptane/air mixtures by considering the role of the cool flame chemistry. It is found that the existence of the cool flame can substantially extend the lean and rich limits of the conventional high-temperature flame, with the extended limits being insensitive to pressure but sensitive to the extent of inert dilution. Different levels of kinetic criteria based on the controlling branching and termination reactions at low temperatures are derived and shown to agree well with the numerically simulated results at normal pressure. Furthermore, the critical temperature of cool flame flammability limits at normal pressure is found to be around 630 K and increases with pressure due to the shift of the NTC regime to higher temperatures. This work is reported in: “Extended flammability limits of n-heptane/air mixtures with cool flames,” by Wenkai Liang and Chung K. Law, Combustion and Flame, Vol. 185, pp. 75-81 (2017). See Bibliography section below. Theory of First-stage Ignition Delay in Hydrocarbon NTC Chemistry: The first-stage ignition delay of n-heptane/air mixtures is computationally studied using detailed mechanism and theoretically studied using eigenvalue analysis of simplified systems. Results show that the delay has a turnover behavior as temperature increases, being dominated by the competition of low-temperature branching and termination channels as well as the competition of forward and reverse reaction channels. As temperature increases to the intermediate range, the termination pathways result in a minimum in the delay, the state of which is theoretically derived.

Task Progress:	The controlling kinetics of the first-stage ignition delay is found to be due to two types of competitions for the key intermediate species, namely the branching-termination competition and the forward-reverse competition. Between the two, the branching-termination competition is found to be the main reason leading to the turnover behavior around the regime transitioning from low to intermediate temperature. Furthermore, among all the branching-termination pairs, the competing channels of QOOH are found to be the most important for the minimum ignition delay. The validity of the derived analytical expression has also been tested and established for the large hydrocarbon fuels of n-octane and iso-octane, demonstrating the generality of the first-stage ignition concept. This work is reported in: “Theory of first-stage ignition delay in hydrocarbon NTC chemistry,” by Wenkai Liang and Chung K. Law, Combustion and Flame, Vol. 188, pp. 162-169 (2018). See Bibliography section below. Direct Numerical Simulation of Turbulent Nonpremixed "Cool" Flames: Applicability of Flamelet Models: “Cool” flames result from the coupling of low-temperature chemistry with molecular transport. These flames have been experimentally and computationally observed under laminar flow conditions but have not been isolated under turbulent flow conditions. A skeletal n-heptane chemical mechanism including low-temperature chemistry is used to conduct two-dimensional Direct Numerical Simulations (DNS) of non-premixed “cool” flames subjected to unsteady, two-dimensional flow initialized from a plane of isotropic turbulence. Like conventional “hot” flames, at high Damköhler numbers, these “cool” flames are found to be adequately described by a steady flamelet model. However, at low Damköhler numbers, both non-premixed and premixed behavior are observed locally. When the combustion is locally non-premixed, these “cool” flames can be described by the unsteady flamelet model only if the correct scalar dissipation rate profile and associated boundary conditions are considered. A comparison of two cases with varying Damköhler numbers showed that steady flamelet models are able to predict the behavior of the “cool” flame when the Damköhler number is sufficiently large, such that chemistry is faster than the flow timescales. In order to understand whether the lower Damköhler number case could be modeled with an unsteady flamelet model, a method was developed that could locally evaluate whether the combustion was locally non-premixed. Local increases in temperature above the steady flamelet model were then attributed to “premixed” effects rather than autoignition toward “hot” flames, which is too slow for the conditions considered. For points identified as locally non-premixed, the unsteady flamelet model was shown to be incorrect when a standard dissipation rate profile was considered. However, when the exact dissipation rate profile was considered, the unsteady flamelet model was shown to be consistent with the DNS data. These analyses indicate that conventional non-premixed combustion models developed for “hot” flames are equally applicable to “cool” flames as long as the underlying assumptions are satisfied. This work is reported in: “Direct numerical simulation of turbulent nonpremixed "cool" flames: applicability of flamelet models,” by Alex G. Novoselov, Chung K. Law and Michael E. Mueller, Proceedings of the Combustion Institute, Vol. 27, pp. 2143-2150 (2019). See Bibliography section below. On Radical-induced Ignition in Combustion Systems: Theoretical developments based on eigenvalue analysis in assessing the state of incipient ignition induced by radical runaway with detailed chemistry of several classical problems are examined. The canonical ignition limits are analyzed, which are conventionally termed explosion limits, of hydrogen-oxygen mixtures, yielding explicit criteria that well reproduce their characteristic Z-shaped response in the pressure-temperature space. Subsequently, the role of hydrogen addition to the explosion limits of carbon monoxide-oxygen and methane-oxygen mixtures is evaluated, demonstrating and quantifying its strong catalytic effect, especially for the carbon monoxide mixtures. The role of low-temperature chemistry (LTC) in the autoignition of large hydrocarbon fuels is discussed, with emphasis on the first-stage ignition delay and the associated negative-temperature coefficient (NTC) phenomena. The analysis is extended to problems of nonhomogeneous ignition in the presence of convective-diffusive transport, using the counterflow as an example, demonstrating the canonical similarity between the homogeneous and nonhomogeneous systems. Specific topics include hydrogen-oxygen ignition limits in homogeneous and nonhomogeneous systems, hydrogen-catalyzed ignition of CO- and CH4-oxygen mixtures, and the low-temperature chemistry in autoignition. The explosion limits of H2-CO-O2 and H2-CH4-O2 mixtures are examined at the next level of fuel complexity. Specifically, it shows that with only minute quantity of hydrogen addition, the explosion limits of the CO-O2 mixtures transition from monotonic to nonmonotonic behavior. The highly sensitive variation of the kinetic parameter characterizing the competition of O radical reacting with H2 and CO well explains such dramatic change. The H2-CH4-O2 mixtures show the transition by hydrogen addition with much less sensitivity. Such a difference is explained based on the competition of O and OH radicals by the branching and termination channels. This work is reported in: “On radical-induced ignition in combustion systems,” by Wenkai Liang and Chung K. Law, Annu. Rev. Chem. Biomol. Eng., Vol. 10, pp. 199-217 (2019). See Bibliography section below.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Novoselov AG, Law CK, Mueller ME. "Direct Numerical Simulation of turbulent nonpremixed “cool” flames: Applicability of flamelet models." Proceedings of the Combustion Institute. 2019;37(2):2143-50. https://doi.org/10.1016/j.proci.2018.06.191 , Jan-2019
Articles in Peer-reviewed Journals	Liang W, Law CK. "On radical-induced ignition in combustion systems." Annu Rev Chem Biomol Eng. 2019 Jun;10:199-217. https://doi.org/10.1146/annurev-chembioeng-060718-030141 , Jun-2019